

A Deep Reinforcement Learning-based Approach for Adaptive Handover Protocols

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Abstract—The use of higher frequencies in mobile communication systems leads to smaller cell sizes, resulting in the deployment of more base stations and an increase in handovers to support user mobility. This can lead to frequent radio link failures and reduced data rates. In this work, we propose a handover optimization method using proximal policy optimization (PPO) to develop an adaptive handover protocol. Our PPO-based agent, implemented in the base stations, is highly adaptive to varying user equipment speeds and outperforms the 3GPP-standardized 5G NR handover procedure in terms of average data rate and radio link failure rate. Additionally, our simulation environment is carefully designed to ensure high accuracy, realistic user movements, and fair benchmarking against the 3GPP handover method.

Index Terms—Handover, Communication Protocols, Mobility Management, Deep Reinforcement Learning

I. INTRODUCTION

The exponential growth in user equipment (UE) connectivity and the rise of high-demand applications, such as autonomous driving, are increasing the operational complexity of cellular networks. This complexity is further amplified by higher frequency bands, which enable faster transmission but lead to smaller cell sizes and dense heterogeneous network (HetNet) architectures. These multi-tier networks, with a mix of macro and micro base stations (BSs) require highly reliable and adaptive handover (HO) protocols to maintain seamless connectivity. Traditional network optimization methods based on static configurations and fixed parameters are insufficient to meet the dynamic needs of modern networks.

With increased user mobility and cell density, HO events become more frequent, especially at cell boundaries where signal strength fluctuates. This increase in HO results in more ping-pongs (PPs) and radio link failures (RLFs), leading to service interruptions and rate losses. PPs, occurring when UEs oscillate between neighboring cells, increase the HO execution time and degrade quality of service (QoS). The connectivity is further affected by RLFs, often due to too early or delayed HOs. Thus, good timing of HOs is essential, but optimal protocol parameters vary widely depending on the network topology, UE behavior, and mobility patterns, making it challenging to optimize them for all UEs in traditional event-based protocols.

To address these challenges, there is a need for protocols that can dynamically adapt to varying QoS demands and environmental conditions. Reinforcement learning (RL) represents

a promising approach that enables protocol optimization based on feedback from the network. RL-based HO protocols can optimize timing and HO decision-making, adapting to UE mobility and QoS requirements.

A. Related work

The optimization of HO protocols in cellular networks has been an active research area, especially with the shift towards dense HetNets where high UE mobility and smaller cell sizes increase the frequency of HOs. Mobility management in LTE and 5G NR, standardized by 3GPP, relies on an event-based protocol with fixed thresholds [1]. This method, however, is limited in adapting to varying conditions in modern networks, often leading to suboptimal performance [2].

Adaptive HO algorithms have been proposed to address these limitations. For instance, context-aware HO mechanisms consider user speed to adjust HO parameters, aiming to maximize UE capacity [3], [4]. These approaches rely on defined thresholds and are generally tailored for specific environments, limiting their adaptability. Machine learning approaches offer more flexible solutions that adapt to changing network conditions. Supervised learning has been used to predict optimal HOs based on historical data [5]. However, these methods rely heavily on data quality and struggle to adapt to dynamic network changes.

In contrast, RL provides a framework that learns by interacting with the environment, eliminating labeled datasets. In HO optimization, RL algorithms show promise in reducing PPs and RLFs by learning adaptive policies that adjust HO timing based on network conditions [6]. Q-learning-based HO methods have demonstrated gains by taking actions based on the Reference Signal Received Power (RSRP) [7], but they face scalability issues in continuous state spaces.

Deep reinforcement learning (DRL) has led to more scalable solutions for HO optimization. DRL methods, such as deep Q-Networks and proximal policy optimization (PPO), have addressed HO decisions in dense networks with complex decision spaces [8], [9]. By using neural networks (NNs) to approximate optimal policies, these methods better handle variations in UE mobility and network conditions. DRL-based HO protocols reduce PPs and RLFs by optimizing HO timing and cell selection based on QoS requirements and mobility predictions. Multi-agent DRL approaches aim to optimize mobility network-wide but introduce higher complexity [10].

B. Contribution

This work investigates the application of RL in optimizing HO protocols within cellular networks. It aims to maximize the average data rate and reduce RLFs by enhancing the flexibility of HO decision-making. Our approach optimizes the timing of HO events, leading to higher signal-to-interference-plus-noise ratio (SINR) during the HO execution, and reducing the probability of RLFs due to handover failures (HOFs).

Our results demonstrate that DRL-based HO protocols offer significant potential to improve the reliability, efficiency, and mobility-aware behavior required in next-generation cellular networks, especially in high-mobility environments. The results are based on simulations modeling realistic UE mobility patterns and network conditions, providing a comprehensive testbed for research on HO optimization. To support further research and reproducibility, we make the source code, datasets, and trained models publicly available¹.

II. PRELIMINARIES

A. Handover Procedure within 5G NR

Similar to previous 3GPP releases, mobility in 5G NR is managed by an event-driven HO procedure [1]. The procedure is based on RSRP measurements of the serving cell and detected neighboring cells. For intra-frequency handovers, as considered in this work, the handover decision is usually based on *Event A3*. The event is triggered when the following condition is met, according to [1, p. 260]:

$$M_n + \text{Off}_n + \text{Off}_{\text{cn}} - \text{Hys} > M_p + \text{Off}_p + \text{Off}_{\text{cp}} + \text{Off}, \quad (1)$$

where M_n and M_p are measurements of the neighboring and serving cell, neighboring cell-specific offsets Off_n , and Off_{cn} , serving cell offsets Off_p , Off_{cp} , Off , and a hysteresis Hys .

When Event A3 is triggered, the UE starts sending measurement reports to the serving BS, i.e. gNodeB, which then makes the HO decision and initiates the following HO preparation and finally sends the HO command to the UE, to start the HO execution. To avoid excessive HOs, the UE only starts sending measurement reports after a certain waiting time, called time-to-trigger (TTT). If the RSRP of the serving cell recovers before the HO command is sent and exceeds a defined threshold, according to the leaving condition [1, p. 260], the HO can be aborted. Too fast changes between the entering and leaving condition of Event A3 are reduced by the hysteresis in Eq. (1). The offsets, TTT, and the hysteresis are adjustable parameters configured by the serving BS. Network operators must optimize these parameters to prevent common HO issues, such as RLF caused by the UE being out-of-synchronization due to poor SINR or RSRP, and PP events, where the UE switches too often back to the previous BS after a HO.

¹Source code, model, and datasets publicly available at: <https://github.com/kit-cel/HandoverOptimDRL>

B. Background on Reinforcement Learning

In RL, an agent interacts with an environment to maximize cumulative rewards over time. At each time step t , the agent observes the state $s_t \in \mathcal{S}$, takes an action $a_t \in \mathcal{A}$ according to its policy π , which initiates the transition to the next state s_{t+1} and receives a reward $r(s_t, a_t) = r_t$. The policy $\pi(a|s)$ defines the probability of selecting action a given state s . In deep RL, the policy is represented by a NN, called policy network, with parameters θ , denoted as π_θ . The goal is to find an optimal policy π^* that maximizes the expected cumulative reward $G_t = \mathbb{E}_\pi [\sum_{k=0}^{\infty} \gamma^k r_{t+k+1}]$, where $\gamma \in [0, 1)$ is the discount factor weighting future rewards.

Policy gradient methods are widely used to find the optimal policy π^* . The actor-critic framework uses two separate NNs: an actor network, representing the policy π_θ , and a critic network v_ϕ , which estimates the expected return G_t from state s_t , known as value function $V_t^\pi(s)$ [11]:

$$V_t^\pi(s) = \mathbb{E}_\pi [G_t | s_t] = \mathbb{E}_\pi \left[r_t + \sum_{k=1}^{\infty} \gamma^k r_{t+k+1} \middle| s_t \right].$$

Policy gradient methods directly optimize the policy by adjusting parameters θ through gradient ascent on the objective function. The gradient with respect to θ can be estimated as:

$$\hat{g} = \hat{\mathbb{E}}_t [\nabla_\theta \log \pi_\theta(a_t | s_t) \hat{A}_t], \quad (2)$$

where $\hat{A}_t$ is an estimate of the advantage function based on V_ϕ [12]. The generalized advantage estimation [13] can be used to estimate $\hat{A}_t$, see Fig-1.

PPO improves the performance by constraining policy updates to prevent drastic changes [12]. This is achieved by optimizing a truncated, i.e., proximal, objective function and thereby improves training stability:

$$L^{\text{Clip}}(\theta) = \hat{\mathbb{E}}_t [\min(\psi_t(\theta) \hat{A}_t, \text{clip}(\psi_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)], \quad (3)$$

where $\psi_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$ is the probability ratio between new and old policies, and ϵ is a hyperparameter that limits policy updates. The critic minimizes the mean squared error between the estimated value $V_\phi(s)$ and a target value $V_{\text{target}} = r_t + \gamma V_\phi(s_{t+1})$ [12], as shown in Fig. 1:

$$L^{\text{Critic}}(\phi) = \mathbb{E}_t [(V_\phi(s_t) - V_{\text{target}})^2]. \quad (4)$$

The policy and value network are optimized using Adam [14] with learning rate α .

III. SYSTEM MODEL

A. The Environment

The considered scenario is a dense urban 5G NR network with a set of N macro BSs, denoted as $\mathcal{B} = \{b_0, \dots, b_{N-1}\}$, and UEs moving randomly through the simulated area at different speeds. The BSs positions are based on real locations from publicly available network information, as shown in Fig. 2. All BSs and UEs use the IMT-2000 (n1) band at 2.1 GHz for data transmission.

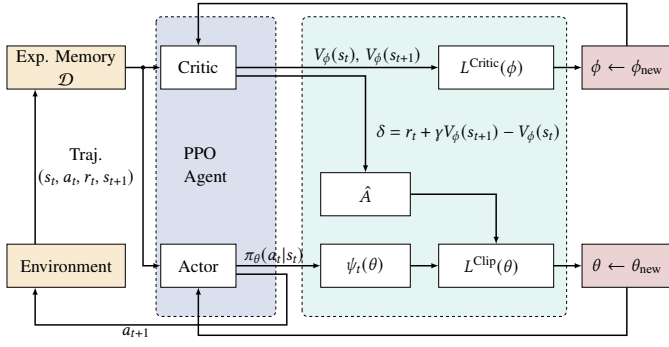


Fig. 1. The centralized DRL agent includes a proximal policy optimization actor-critic unit (violet), the environment and experience buffer (orange), loss calculation (green), and optimizer (red).

We use the *Vienna 5G System Level Simulator* [15] with environmental data such as building floor plans and street systems obtained by OpenStreetMap. The region of interest² is near the city center and includes pedestrian zones, side roads, and main streets covering diverse mobility scenarios. UE speeds range from 3 km/h to 50 km/h, representing pedestrians, cyclists, and cars. To test adaptability, HO protocols in Sec. V will also be evaluated at higher speeds up to 90 km/h.

To generate realistic movement patterns and keep users on roads, we use *Simulation of Urban Mobility* [16]. GPX tracks for UEs are sampled every $\Delta t = 10$ ms. The temporal resolution Δt is chosen such that handover-related procedures defined in Sec. II-A and Sec. III-B can be correctly resolved and simulated. These tracks are then used within the Vienna 5G Simulator to calculate the RSRP and SINR values for each UE-BS connection at each time step, based on the UMa 5G channel model [17]. Line-of-sight and shadowing effects are derived from 3D building data. The generated RSRP and SINR are assumed to be layer 1 filtered to reduce fast-fading effects. Additionally, layer 3 filtering is applied according to [1].

B. Simulation of Handover, PP, and RLF Events

Accurately simulating HOs, PPs, and RLFs is essential for comparing 3GPP mobility management with the DRL-based approach. HOs are modeled with a preparation phase followed by an execution phase. During preparation, HOs can be aborted if signal conditions change. The HO command is received by the UE at the end of the HO preparation, after which the UE connects to the target BS. The following definitions, based on [18], are used to represent RLFs, HOFs, and PPs in the simulation:

Definition 1 (Radio Link Failure): An RLF occurs if the UE experiences out-of-sync conditions, i.e., $\text{SINR} < Q_{\text{out}}$. After N310 consecutive out-of-sync indicators from lower layers, timer T310 will be started. If T310 expires before recovery, i.e., N311 in-sync indicators ($\text{SINR} > Q_{\text{in}}$) are received, an RLF is declared.

²Coordinates: 49°00'17.6"N 8°22'38.3"E – 49°00'40.3"N 8°23'44.2"E

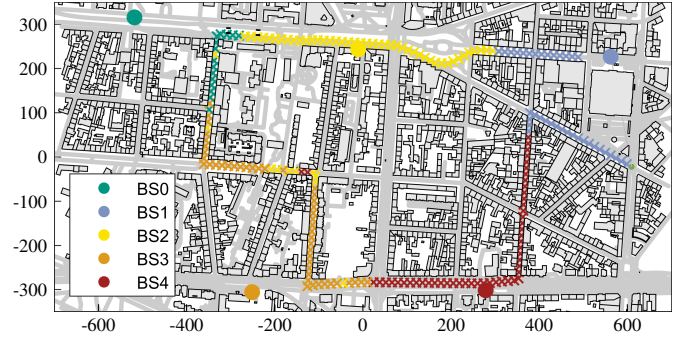


Fig. 2. Simulated region with areas of different speed limits and pedestrian zones. BSs are indicated by circles and crosses mark the path of an UE in the color of the serving BS. The axes indicate the distance in meters.

Definition 2 (Handover Failure): An HOF triggers an RLF if the HO command is received while timer T310 is running, as shown in Fig. 3. Additionally, if T310 expires before the HO execution starts, an HOF and RLF occurs.

Definition 3 (Ping-pong Handover): A PP is defined as a HO where the UE switches to a new BS and then returns to the previous BS within a period shorter than a specified minimum-time-of-stay (MTS).

Table I shows the parameters for modeling PPs, RLFs, and the 3GPP event-based HO procedure [4] [18].

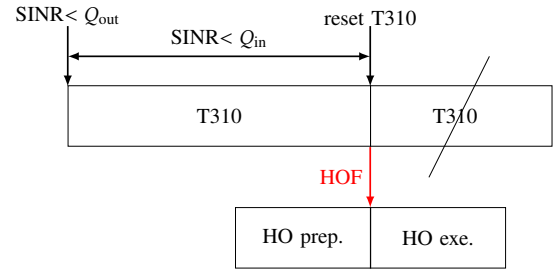


Fig. 3. HOF triggered after HO preparation, causing an RLF.

IV. RL PROBLEM FORMULATION AND OBJECTIVE

The goal of the PPO-based agent is to maximize the average rate of the UEs. We model the problem based on Sec. II, such that the agent is localized in each BS and receives measurements from the UEs on which the agent's decisions are based. It is assumed that the agent is trained in advance (before deployment) to be ready for operation, but it can be further optimized dynamically by updating the network parameters.

A. State

The state at any time t , denoted by s_t , is represented by a vector containing important information about the network conditions and the status of the UE. The state s_t includes a one-hot encoded vector $s_t^{\text{BS}} = [0, \dots, 1, \dots, 0] \in \mathbb{R}_2^N$ to indicate the current serving BS, where the position corresponding to the serving BS among N available BSs is 1, while all other

TABLE I
PARAMETERS USED FOR 3GPP PROTOCOL & RLF DETECTION

Parameter	Value
Q_{in}	-6 dB
Q_{out}	-8 dB
A3 hysteresis (Hys)	1 dB
A3 Offset (Off)	[0, 1, 2] dB
A3 TTT	[40, 80, 160] ms
HO preparation time	50 ms
HO execution time	40 ms
MTS	1000 ms
T310	1000 ms
Avg. RLF recovery	200 ms
Counter N310	10
Counter N311	3

entries are 0, allowing the agent to identify the serving BS. The i -th entry of $\mathbf{s}_t$, denoted $s_t(i)$, is defined as follows:

$$s_t^{BS}(i) = \begin{cases} 1, & \text{if } b_i \in \mathcal{B} \text{ is the serving BS} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

To add information about the signal quality from each BS, we include the SINR. The SINR is clipped to $[-10, 10]$ dB according to the empirical cumulative distribution function (ECDF) in the time period around the handover to focus on the HO decision-making. This speeds up training because there is no need for HOs during phases where the serving BS is clearly the best BS. The clipped SINR values are then scaled to $[0, 1]$, providing a normalized indication of connectivity quality in the same scale as the one-hot encoded BS indicator vector $\mathbf{s}_t^{BS}$:

$$\mathbf{s}_t^{SINR} = [q_0, q_1, \dots, q_{N-1}], \quad (6)$$

where each $q_i \in [0, 1]$ represents the clipped and scaled SINR of BS $b_i \in \mathcal{B}$. This vector helps the agent evaluate the relative connection quality across potential target BSs for HOs.

Furthermore, to discourage PPs, the state vector includes a binary indicator flag. This indicator is set to 1 if the elapsed time since the last HO, $t - t_{HO}$, is less than a specified MTS, and 0 otherwise:

$$s_t^{PP} = \begin{cases} 1, & \text{if } t - t_{HO} < \text{MTS} \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where t_{HO} denotes the last HO time. Including this PP indicator aids the agent in stabilizing the UE connection by avoiding oscillations between BSs. Thus, the full state vector $\mathbf{s}_t$ at time t and of size $2N + a$ is constructed as:

$$\mathbf{s}_t = [\mathbf{s}_t^{BS}, \mathbf{s}_t^{SINR}, \mathbf{s}_t^{PP}] \in \mathcal{S}. \quad (8)$$

B. Action

At each time step t , the agent chooses an action $a_t \in \mathcal{A} = \{0, \dots, N-1\}$ that corresponds to the index of a BS.

The action either initiates a HO to a neighboring BS or the UE remains connected to the serving BS. If the agent selects a different target BS while a HO is already in preparation, the current preparation is aborted, the target BS is updated, and the HO process restarts.

C. Reward

The reward function $r_t(\mathbf{s}_t, a_t)$ is designed to encourage stable, high-quality connections and to avoid PPs and RLFs by penalizing them. The reward consists of three components:

$$r_t(\mathbf{s}_t, a_t) = r_{SINR}(\mathbf{s}_t, a_t) + r_{PP}(\mathbf{s}_t, a_t) + r_{RLF}(\mathbf{s}_t).$$

Each component relates to a specific aspect of HO performance, where $r_{SINR}(\mathbf{s}_t, a_t)$ rewarding high SINR values and encouraging the agent to select BSs that provide better connectivity, with a bonus $C \in \mathbb{R}$ for the currently best BS:

$$r_{SINR}(\mathbf{s}_t, a_t) = \begin{cases} s_t^{SINR}(i) + C, & \text{if } i = \arg \max_{i \in \mathcal{B}} s_t^{SINR}(i) \\ s_t^{SINR}(i), & \text{otherwise.} \end{cases} \quad (9)$$

To prevent PPs, the second component, $r_{PP}(\mathbf{s}_t, a_t)$, applies a penalty when a PP occurs:

$$r_{PP}(\mathbf{s}_t, a_t) = \begin{cases} -C, & \text{if PP detected} \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Similar to the predecessor, $r_{RLF}(\mathbf{s}_t, a_t)$ penalizes the agent for allowing the UE to experience out-of-sync conditions or RLFs, emphasizing the importance of avoiding service disruptions:

$$r_{RLF}(\mathbf{s}_t, a_t) = \begin{cases} -C, & \text{if SINR indicates out-of-sync} \\ -2C, & \text{if RLF detected} \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

D. Objective

In this framework, the PPO agent aims to maximize the average data rate by learning an optimal policy for HO decisions. The average rate $\bar{R}$ is calculated as:

$$\bar{R} = \frac{1}{T} \sum_{t=0}^{T-1} B \log_2(1 + \text{SINR}_{b(t)}(t)), \quad (12)$$

where B represents the bandwidth, T is the total number of time steps, and $\text{SINR}_{b(t)}(t)$ is the SINR at the serving BS $b(t) \in \mathcal{B}$ at time t .

To provide an upper bound for the achievable rate, we define $\bar{R}_{\max}$ as the maximum possible data rate if the HO execution time was zero, allowing the UE to always connect to the BS with the highest SINR. This idealized rate is given by:

$$\bar{R}_{\max} = \frac{1}{T} \sum_{t=0}^{T-1} \max_{b \in \mathcal{B}} B \log_2(1 + \text{SINR}_b(t)). \quad (13)$$

The agent's performance is then assessed through the relative average rate $\Gamma_R \in [0, 1]$, defined as:

$$\Gamma_R = \frac{\bar{R}}{\bar{R}_{\max}}. \quad (14)$$

This metric, Γ_R , allows us to evaluate how closely the agent's policy approximates the optimal rate in the absence of HO delays and to compare it with the 3GPP HO procedure as a benchmark.

In addition to maximizing data rate, a secondary objective is to minimize RLFs by minimizing HOFs, thereby reducing service interruptions and improving connection stability. By balancing these objectives, the agent seeks to optimize both the quality and reliability of the UE's connection within the network.

E. Training Process

In each episode, a random training dataset is selected to support training. To further enhance generalization and prevent the agent from becoming reliant on a specific BS, the BS indices $b \in \mathcal{B}$, are shuffled after each episode. The training process consists of two major phases. In the first phase, the episode terminates only if a RLF is detected or the maximum number of episode time steps T is reached, allowing the agent to focus on learning optimal HO strategies. In the second phase, episodes are also terminated by a PP event, introducing a penalty for excessive HOs and encouraging more stable connections.

The agent is implemented using PPO from the Stable-Baselines3 library [19]. Tab. II provides training-specific parameters determined by hyperparameter optimization.

TABLE II
TRAINING & PPO AGENT RELATED PARAMETERS.

Parameter	Value
C (reward/penalty)	0.95
Learning rate α	$5 \cdot 10^{-5}$
NN architecture	[64, 128, 64]
PPO ent _{coef}	0.1

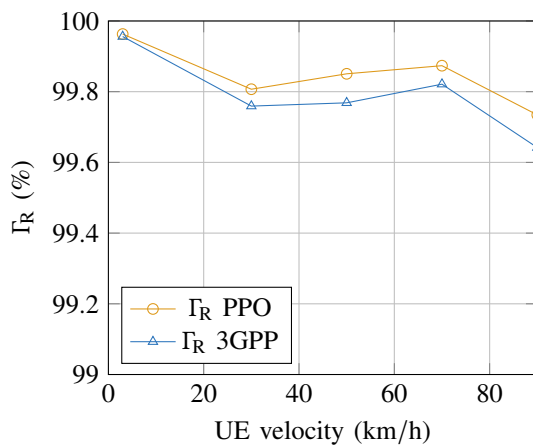


Fig. 4. Achieved relative average rate Γ_R of the 3GPP and PPO-based HO protocol for different UE velocities.

V. NUMERICAL RESULTS

The performance of both the 3GPP HO procedure and the proximal policy optimization agent was evaluated on datasets generated using the setup described in Sec. III. UEs moving at speeds of up to 50 km/h were used, with parameters provided in Tab. I and Tab. II to optimize the parameters of the 3GPP HO procedure and to train the PPO agent. The 3GPP HO protocol was optimized to perform well on average on the training data sets. However, no additional optimizations for individual UE speeds were made for either protocol, as the adaptivity of the protocols is being investigated.

Both protocols were tested on randomly generated datasets, with additional tests using UEs traveling at speeds up to 90 km/h to evaluate adaptability to varying environmental conditions. During testing, the PPO agent was not terminated upon experiencing PP events or during the RLF recovery. Actions during these periods are ignored because the agent cannot perform actions during this time, i. e., the actions do not change the state of the environment. However, as previously described, the HO preparation could be modified or aborted if necessary.

Fig. 4 presents the relative average rate Γ_R as calculated in Eq. 14 for both the reference protocol and the PPO agent across varying UE speeds, averaged over all test datasets. The results show that our PPO-based approach achieves slightly better relative average rates than the reference protocol. In particular, the performance gap to the theoretical upper bound remains relatively small. This is mostly because UEs are connected to the optimal BS most of the time, and the relative duration required for HO execution or RLF recovery is minimal. However, as UE speeds increase, this gap also increases, which is expected since higher speeds require more frequent HOs, where precise timing is more critical.

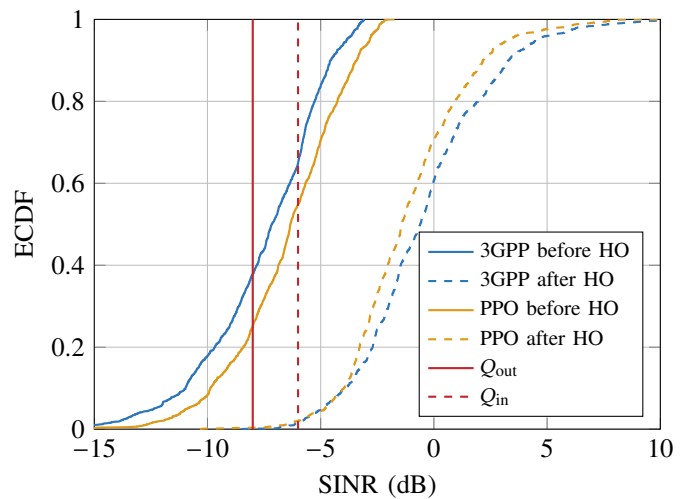


Fig. 5. ECDF of SINR before (solid) and after (dashed) handover execution under 3GPP and PPO policies. Red lines indicate Q_{out} (solid) and Q_{in} (dashed) SINR QoS thresholds.

An investigation of HO timing provides further insights into HO performance. Fig. 5 shows that the HO timing achieved by the DRL agent is better than the timing of the 3GPP reference. The figure displays the ECDF of the SINR at the start (solid lines) and end (dashed lines) of the HO execution, representing the SINR of the serving BS upon receipt of the HO command and the SINR of the target BS at the HO completion. Additionally, QoS thresholds for Q_{out} and Q_{in} , important for RLF detection, are marked as solid/dashed red lines. The results show that the probability $P(\text{SINR} \leq Q_{out})$ at the time of HO is reduced by more than 30% compared to the reference, leading to a decrease in HOFs, as shown in Fig. 6.

It is also worth mentioning that the SINR of the PPO agent post-HO completion is, on average, slightly lower than that of the reference protocol, due to the agent's earlier HO timing. However, the difference in probabilities within the RLF-relevant SINR range is negligible, underscoring the improved HO timing. The lower HOF probability in Fig. 6 confirms the effectiveness in reducing the RLFs frequency caused by HOs.

Fig. 6 compares the average HOF and PP probability of the 3GPP protocol and the PPO agent for different UE speeds. The PPO agent outperforms the 3GPP protocol in terms of HOF probability, while the PP probability is at a similar level for both protocols. The PP rate of the 3GPP protocol is relatively high due to the chosen parameter TTT of 40 ms, but leads to a higher average rate than larger TTTs.

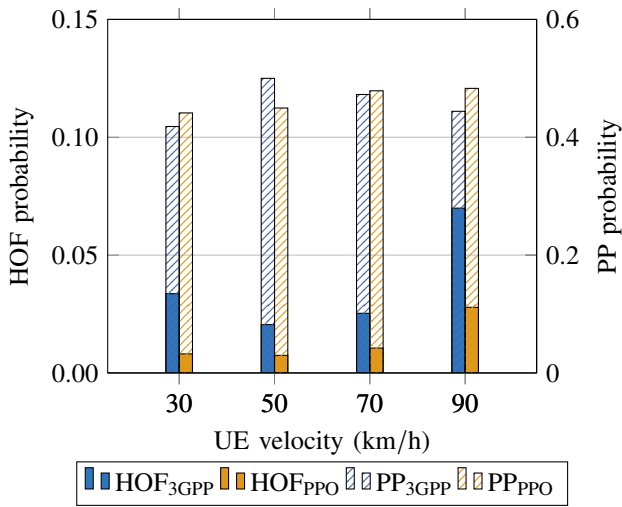


Fig. 6. HOF and PP probability.

VI. CONCLUSION

This paper has presented a reinforcement learning-based approach using proximal policy optimization to enhance handover protocols in dense cellular networks. By focusing on optimal handover timing and decisions, the proposed PPO-based agent adapts dynamically to various user mobility patterns and network conditions. Our results demonstrate that the adaptive approach achieves at least the same average data rates, but reduces RLFs compared to the 3GPP-standardized 5G NR HO protocol, especially in high-mobility scenarios. The

adaptability of DRL-based protocols offers promising potential for next-generation cellular networks towards more reliable and efficient connectivity in HetNets. This work contributes to the growing field of mobility management optimization, with publicly available source code and datasets for further research and development.

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